A Symbol-Based Passband Doppler Tracking and Compensation Algorithm for Underwater Acoustic DSSS Communications

Dajun Sun, Xiaoping Hong, Hongyu Cui, Lu Liu

***Abstract*—In this paper, we consider the problem of Doppler tracking and compensation for a direct sequence spread spectrum (DSSS) signal in underwater acoustic (UWA) communication. Since the dynamic property of the UWA channel and the long duration of DSSS signals result in significant Doppler spread that severely distorts the propagated signal, Doppler tracking and compensation are required. Based on the ultra-wideband property of UWA signal, the Doppler spread not only results in the frequency shift, but also changes the signal duration. Therefore, the accurate estimation of the signal expansion/compression in the time domain can reflect the Doppler spread. Accordingly, we present a Doppler tracking and compensation algorithm for a DSSS signal operating on the correlation output of passband signal at a symbol-by-symbol basis. Note that the carrier frequency of UWA communication is around several kilohertz, and thus the time delay estimation can be performed on the passband to improve the accuracy. Furthermore, the prior information of Doppler limit is used to refine the resolution of delay estimation and achieve sequential estimation. To compensate the correlation magnitude distortion induced by the velocity variation, the local reference signal is selected adaptively based on the filtered Doppler factor. Simulation results demonstrate that the proposed passband Doppler tracking algorithm achieves a superior performance compared with the conventional receiver.**

***Keywords*—UWA communications, DSSS, passband Doppler tracking and compensation, three-point interpolation**

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I. Introduction

Since the dynamic property of underwater acoustic (UWA) channel and the slow propagation speed of acoustic waves, the motion-induced and environment-induced Doppler distortion of acoustic signal is extremely severe[1,2]. The speed of acoustic waves is very low compared with the speed of electro-magnetic waves, and thus the Doppler effect is described not only as frequency shifting but also as time scaling (expansion or compression) of the signal waveform underwater[3,4]. Furthermore, the available bandwidth of UWA communication is quite limited, i.e., less than 20 kHz in general, and the packet duration of UWA communications is thus long, i.e., from several seconds to more than ten seconds[5]. Accordingly, longer transmission packet experiences more severe Doppler distortion, especially for the direct sequence spread spectrum (DSSS) transmission scheme. Therefore, the Doppler tracking and compensation for the DSSS signal is quite necessary for UWA communication.

Among the researches over Doppler estimation of UWA communication, the widely used method is based on computing the cross-ambiguity function (CAF)[6]. This method utilizes multi-channel correlation integrated with different Doppler factors to achieve coarse Doppler estimation. The Doppler estimation accuracy is limited by the type and the duration of communication signal. It could be extended to dynamic Doppler tracking for a DSSS signal on a symbol basis. However, the CAF method is computationally intensive due to a large number of Doppler estimation scales, and the Doppler estimation is not accurate enough for compensation. Therefore, it is typically applied to coarse Doppler estimation as a pre-processor.

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More researches over Doppler estimation of UWA communication focus on the block estimation, including measuring the time difference of Doppler-insensitive pulse signal pair between packet head and tail[2,7], and Doppler scale estimation based on two adjacent and identical symbols in the frequency domain[8]. However, these methods assume the Doppler scale is nearly constant over the data packet and only the initial or the average Doppler scale is estimated, which is not valid for many scenarios, especially for long-duration DSSS signals. Doppler estimation considering the dynamic change within a data packet is addressed in later researches. The first widely used method is the adaptive decision feedback equalization (DFE) integrated with phase locked loop (PLL)[9], which works well when residual Doppler is small. The improvements over this method are done by SHARIF in Refs. [4,10], who addressed a close lock loop to estimate the Doppler factor and resample the baseband signal adaptively, and SINGER in Refs. [11,12], who emphasized the broadband Doppler tracking and compensation, and iterative processing to improve the system performance. However, these methods are based on the decision of transmitted symbols, which is ambiguous due to the constellation and sensitive to a decoding error. Recently, ZAKHAROV proposed a superimposed data structure to dynamically track the Doppler scales in Ref. [13]. The kernel is a multibranch autocorrelation (MBA) estimator, which offers an accurate estimate at a low computation load[1,14]. Similar ideas appeared in Ref. [15], in which the pilot signal in the form of continuous wave was used to offer instant Doppler estimate symbol by symbol. However, this kind of methods suffer performance loss in terms of signal-to-noise (SNR) due to the operation of autocorrelation.

In this paper, we propose a novel Doppler tracking and compensation algorithm for DSSS signals operating on cross-correlation output of the passband signal at a symbol-by-symbol basis. Note that the carrier frequency of UWA communications is around several kilohertz, and thus the computational load of passband operation is acceptable and a higher accuracy of time delay estimation can be achieved by the passband correlation[16]. The fractional time delay estimation is based on the three-point interpolation method by the cosine waveform approximation[17]. The time difference of correlation peaks between two adjacent symbols is calculated accordingly, which is related to the Doppler factor on the symbol basis. The Doppler estimation can be filtered sequentially. Furthermore, the prior information of Doppler limit is used to refine the resolution of delay estimation and achieve sequential estimation. To compensate the correlation magnitude distortion induced by the velocity variation, the local reference signal is selected adaptively based on the filtered Doppler factor.

To summarize, the contribution of this paper is to propose a Doppler tracking and compensation algorithm suitable for UWA signals. The Doppler estimation is accurate and updated on the symbol basis, and then the phase rotation and the correlation loss can also be compensated to improve the performance with reasonable complexity. This paper is organized as follows. Section II presents the fundamentals of system models. The detailed description of the proposed algorithm is followed in section III. Simulation results are shown in section IV and conclusions are summarized in section V.

II. System Description

A. Fundamentals of DSSS Signals

We consider the problem of Doppler tracking for DSSS signals in this paper. The key of the proposed algorithm is to measure the time difference of two adjacent symbols. To simplify our discussion, a differential BPSK (DBPSK) modulation scheme is considered for the DSSS signal as

$$b_n = a_n b_{n-1}, \tag{1}$$

where $\{a_n\}$ denotes the information symbols, and $\{b_n\}$ denotes the differential encoded symbols with a length of N. The nth information symbol is decoded based on the two adjacent demodulated symbols as

$$\hat{a}_n = \hat{b}_n \hat{b}_{n-1}^*. \tag{2}$$

Then, the transmitting signal is presented as

$$s(t) = \Re\left\{\sum_{n=0}^{N-1} b[n] \sum_{m=0}^{L-1} c[m]\, g(t - nT_b - mT_c) \mathrm{e}^{\mathrm{j}2\pi f_c t}\right\}, \tag{3}$$

where T_b denotes the duration of a symbol and the sequences $\{c[m]\}$ denote the spread sequences with a length of L and a chip interval of T_c. f_c denotes the carrier frequency, and $g(t)$ denotes the transmit pulse shaping filter, e.g., a rectangle window in this paper

$$g(t) = \begin{cases} 1, & 0 < t \leqslant T_c, \\ 0, & \text{others}. \end{cases} \tag{4}$$

Let $s_n(t)$ represent the transmitting signal at nth symbol, and it is expressed as

$$s_n(t) = \Re\left\{b[n] \sum_{m=0}^{L-1} c[m] g(t - nT_b - mT_c)\, \mathrm{e}^{\mathrm{j}2\pi f_c t}\right\}, \tag{5}$$

where $(n-1)T_b < t \leqslant nT_b$. Therefore, the transmitting signal in (3) can be further rewritten as

$$s(t) = \sum_{n=0}^{N-1} s_n(t), \tag{6}$$

where the first symbol, $s_0(t)$, is used to synchronize the whole frame signal.

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B. Time-Varying Channel

The propagated signals suffer both multipath and Doppler distortions in UWA channels. Because a DSSS signal has a good auto-correlation property, each multipath can be distinguished specifically. It means that the influence of multipath distortions can be neglected for a DSSS transmission scheme. Without loss of generality, we focus on the case of line of sight with a time-varying property induced by Doppler distortion. Other reflected components can also be estimated and compensated by the proposed algorithm. According to Ref. [12], the receiving signal with Doppler distortion is given as

$$r(t) = s\left(t - \frac{d(t)}{c}\right) + n(t), \tag{7}$$

where $s(t)$ and $r(t)$ denote the transmitting and receiving signal, respectively. c denotes the underwater sound speed, and $n(t)$ denotes the additive white Gaussian noise. Moreover, $d(t)$ represents the relative distance of transmitter and receiver versus instant time, and the constant acceleration model is used in this paper as

$$d(t) = d_0 + v_0 t + \frac{1}{2}at^2, \tag{8}$$

where d_0 and v_0 denote the initial distance and speed, respectively, and a denotes the acceleration speed. Actually, more complicated Doppler models can be investigated here, which can be done through defining the motion model of both transmitter and receiver sides. To simplify the discussion, we mainly focus on one-dimension constant acceleration model herein.

To better understand the Doppler effect in UWA channels, we assume that $d_0 = 0$ and $a = 0$, i.e., constant velocity model. Then, we substitute (3) into (7) to further calculate the expression of the receiving signal.

$$r(t) = \Re\left\{\sum_{n=0}^{N-1} b[n] \sum_{m=0}^{L-1} c[m]\, g\left(\left(1 - \frac{v_0}{c}\right)t - nT_b - mT_c\right) \times \mathrm{e}^{\mathrm{j}2\pi f_c(1-v_0/c)t}\right\} + n(t). \tag{9}$$

The above equation shows that the signal duration and carrier frequency change at the same time due to Doppler effect, and these changes are expressed as

$$\begin{cases} \Delta T = \left(1/\left(1 - \dfrac{v_0}{c}\right) - 1\right) NT_b = \dfrac{v_0}{c - v_0} NT_b, \\ \Delta f_c = -\dfrac{v_0}{c} f_c. \end{cases} \tag{10}$$

It shows that when $v_0 > 0$, the signal duration would increase, while the carrier frequency would decrease, and vice versa. It also shows that ΔT is approaching zero when c is relatively large, which is valid in most radio wireless communications. It is not valid in underwater acoustic communication because the underwater sound speed is really slow. Therefore, it is possible to track the phase rotation introduced by the residual Doppler based on delay estimation of two adjacent symbols.

C. Passband Correlation Output

Let $r_n(t)$ represent the receiving signal at the nth symbol, and it is expressed as

$$r_n(t) = \Re\left\{b[n] \sum_{m=0}^{L-1} c[m] g((1-\alpha_n)t - nT_b - mT_c) \times \mathrm{e}^{\mathrm{j}2\pi f_c(1-\alpha_n)t}\right\} + n(t), \tag{11}$$

where $\alpha_n = v_n/c$ denotes the instant Doppler factor for the nth symbol. Note that the Doppler is assumed constant within a symbol here. Then, the receiving signal can be rewritten as

$$r(t) = \sum_{n=0}^{N-1} r_n(t). \tag{12}$$

Let $s_r(t)$ represent the local passband reference signal, and be denoted as (13)

$$s_r(t) = 2 \cdot \sum_{m=0}^{L-1} c[m]\, g(t - mT_c) \mathrm{e}^{\mathrm{j}2\pi f_c t}. \tag{13}$$

Then the cross correlation of the receiving signal and the local passband reference signal at the nth symbol is calculated as

$$R_{nr}(\tau) = \int_{-\infty}^{+\infty} r_n(t) s_r^*(t+\tau)\,\mathrm{d}t = b[n] R_c((1-\alpha_n)\tau - nT_b)\,\mathrm{e}^{\mathrm{j}2\pi(1-\alpha_n)f_c\tau} + \omega(\tau), \tag{14}$$

where $R_c(\tau)$ represents the autocorrelation of transmitting spread sequences convoluted with a rectangle window, and $\omega(\tau)$ denotes the equivalent correlation noise term.

Assume the length of spread sequence is definitely long, and there is no residual Doppler, i.e., $\alpha_n = 0$. Then, the output of cross correlation is corresponding to a baseband signal modulated by the carrier signal, and is given as

$$R_{nr}(\tau) = b(\tau)\,\mathrm{e}^{\mathrm{j}2\pi f_c\tau} + \omega(\tau), \tag{15}$$

where $b(\tau)$ denotes the equivalent baseband signal, which contains the information of transmitting symbols and the other term denotes the equivalent modulated carrier signal.

According to (14) and the definition of $b(\tau)$ in (15), the transmitted symbol can be obtained as

$$\hat{b}[n] = b(\tau_n), \tag{16}$$

where $\tau_n = nT_b$ theoretically, which is valid only under the condition of accurate synchronization and no residual Doppler spread.

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However, the DSSS signal can still be demodulated from the output of cross correlation shown in (14). More specifically, the time index of the nth symbol is given directly from the envelop of the correlation output, i.e.,

$$\hat{\tau}_n = \max_{\tau_n} \left\{ \left| b[n] R_c\left((1-\alpha_n)\tau_n - nT_b\right) \mathrm{e}^{\mathrm{j}2\pi(1-\alpha_n)f_c\tau_n} \right| \right\} = \max_{\tau_n} \left\{ \left| R_c\left((1-\alpha_n)\tau_n - nT_b\right) \right| \right\}. \quad (17)$$

Furthermore, the proposed Doppler tracking algorithm is operated on the output of (14), in which the output is similar to the expression of a passband signal. Note that time delay estimation based on the passband correlation is more accurate than that is based on the baseband correlation[16]. Therefore, we estimate the time delay on the passband basis, and the Doppler estimation is subsequently more accurate.

III. Proposed Doppler Tracking and Compensation Algorithm

After describing the system model in section II, the proposed dynamic Doppler tracking and compensation for DSSS signals, and the complexity analysis is presented in this section. As shown in Fig. 1, the received signal is firstly synchronized. The synchronization is realized through the CAF method based on calculating the CAF with different Doppler factors. In this stage, both time synchronization and coarse Doppler estimation can be implemented. The first local reference signal is generated based on the coarse Doppler estimation result, and then is correlated with the received signal with a duration of two-symbol period. After that, the time difference between the adjacent symbols through a high resolution delay estimator, and the instant Doppler factor is then calculated accordingly. To avoid the instant noise and rapid changes of local reference signal, a first-order low-pass filter is used to smooth the estimate of Doppler factor. And the filtered Doppler factor is used to generate a new local reference signal for the next symbol. Furthermore, a refined symbol is achieved after passband-to-baseband demodulation, baseband phase estimation, and phase rotation compensation.

A. Synchronization and Correlation

As mentioned above, the received signal is firstly synchronized by blindly computing the CAF between the received signal and the local reference signals with different Doppler factors. The first symbol is used to synchronize the whole frame signal. In this stage, both time synchronization and coarse Doppler estimation are implemented. The number and step size of Doppler factor are related to maximum speed of transmitter and receiver, and Doppler limit of a DSSS signal, respectively. Furthermore, the step size of Doppler factor should satisfy that the magnitude loss of correlation output is negligible for the symbol decision. Maximum relative speed of transmitter and receiver is related to application scenario, which is always given. To decrease the computational load of receiver, the local reference signals with different Doppler factors are stored in the memory.

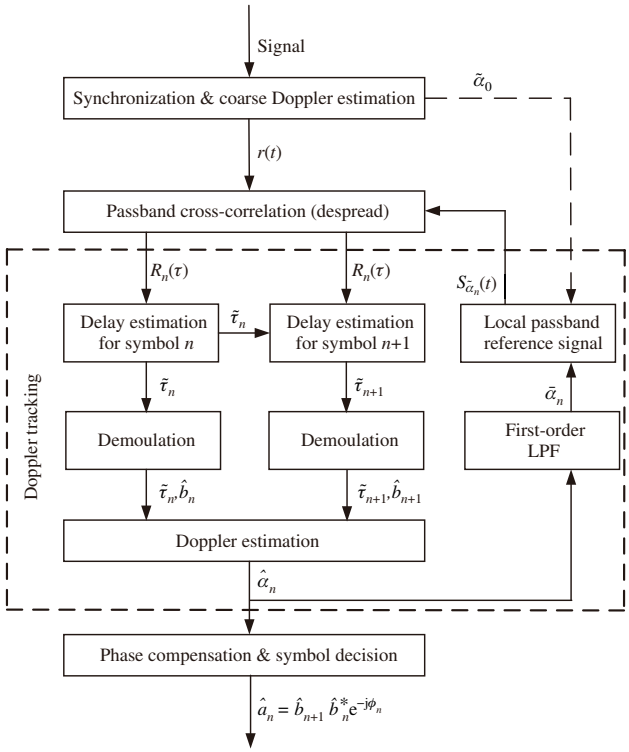

Figure 1 Structure of the proposed dynamic Doppler tracking and compensation algorithm

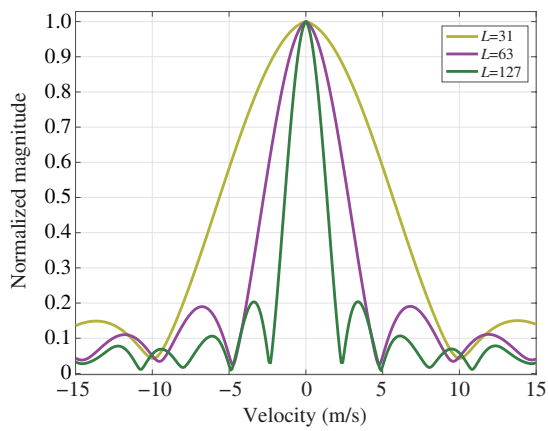

Figure 2 The magnitude loss of correlation peaks caused by the velocity bias between the received signal and local reference signal

The complexity of CAF method is proportional to the number of Doppler factors that are used to generate parallel filters. The number of Doppler factors is related to the Doppler limit of a DSSS signal. Moreover, the local reference signal is required to be selected adaptively based on the instant Doppler

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estimation results, whereas selecting the local reference signal of every symbol is not necessary. Therefore, it is crucial to analyze the Doppler limit to decide the step size of both synchronization and local reference signal generator. According to Ref. [2], the Doppler limit is related to the magnitude loss of the correlation output, i.e., $|R_{0r}(0)|$ in this paper. Fig. 2 shows the magnitude loss of correlation peaks caused by the velocity bias between the received signal and the local reference signal. It is concluded that the Doppler limit decreases with the length of spread sequence.

To better understand the process of synchronization, an example is presented here. The transmitting signal is modulated with a center frequency of 12.5 kHz and a chip rate of 2 500 chips per second (cps). The sample frequency is 100 kHz and the length of spread sequence is 31. The magnitude loss with different Doppler spread factors is given in Fig. 2. To avoid magnitude loss, a Doppler step of around 1.2 m/s is set in this case. The maximum magnitude loss of the signal is less than 0.5 dB. Moreover, the maximum relative speed of the transmitter and receiver is assumed to be no more than 6 m/s. A total of 11 local reference signals are stored in memory.

After synchronizing the desired signal, the dynamic Doppler tracking is operated on a symbol-by-symbol basis. In each processing period, the signal with a duration of two symbol intervals is contained. The signal is correlated with a local passband reference signal, and is used to estimate the time difference between two adjacent symbols.

Assume the Doppler factor is constant over a symbol interval and denoted as α_n, i.e., the ratio of instant velocity to the sound speed, the received signal at the nth information symbol can be modeled as

$$r_n(t) = \Re\left\{\sum_{k=n}^{n+1} b[k] \sum_{m=0}^{L-1} c[m]\, g((1-\alpha_k)t - nT_b - mT_c) \times e^{j2\pi f_c(1-\alpha_k)t}\right\} + n(t). \tag{18}$$

The local reference signal is generated by the local reference signal, and it is expressed as

$$s_{\tilde{\alpha}_n}(t) = \sum_{m=0}^{L-1} c[m]\, g((1-\tilde{\alpha}_n)t - mT_c) e^{j2\pi f_c(1-\tilde{\alpha}_n)t}, \tag{19}$$

where $\tilde{\alpha}_n$ denotes the instant selected Doppler factor, and it is corresponding to a special reference signal.

Then, the passband cross-correlation output is approximately expressed as

$$R_n(\tau) \approx \sum_{k=n}^{n+1} b(k) R_c(\tau - \tau_k) e^{j2\pi f_c(1+\alpha_k-\tilde{\alpha}_n)(\tau-\tau_k)}. \tag{20}$$

B. Delay Estimation and Doppler Estimation

The fraction time delay estimation based on the passband correlation is achieved by a modified three-point interpolation. The interpolation method is originated from delay estimation for pulse signal in the field of target positioning[17], which offers a higher resolution beyond sampling intervals. In this part, we extend the delay estimator to estimate the time difference between two adjacent symbols, which is corresponding to the instant Doppler factor. The Doppler estimation is independent to symbol decision in the receiver, therefore, there is no error propagation when it comes to the symbol decision part.

The passband correlation near the peak value can be approximated by the cosine function, i.e., $A\cos(2\pi f_c \tau)$, in which three unknown parameters can be estimated by the least square criterion. The peak value and its two adjacent values of the sampled signal are selected to estimate the interpolation function, and the fractional time delay estimation can be obtained by solving the derivation of the interpolation function equaling zero. To be more specific, the delay estimator can be simplified as

$$\tilde{\tau} = f_{\mathrm{TD}}(R(\tau), \hat{\tau}, \Delta\tau), \tag{21}$$

where $f_{\mathrm{TD}}(\cdot)$ denotes the three-point interpolation method based on cosine waveform approximation, $\hat{\tau}$ denotes the time delay index of the peak point of correlation envelop, and it is calculated as

$$\hat{\tau} = \max_{\tau} |R_c(\tau)|, \tag{22}$$

and $\Delta\tau$ denotes the phase limit which satisfies $|f_c \Delta\tau| < \pi$.

Now considering the sequential delay estimation of the passband correlation for the continuous DSSS signal, the instant Doppler factor and corresponding instant speed can be estimated.

Recalling (20), it is noted that the real part of correlation output can be formulated as the form of equivalent baseband signal modulated by a cosine waveform. Therefore, the delay estimator suitable for pulse signal is applicable here with a little modification. The equivalent estimation for the continuous DSSS signal is described as

$$\begin{cases} \tilde{\tau}_n = f_{\mathrm{TD}}(|\Re\{R_n(\tau)\}|, \hat{\tau}_n, \Delta\tau), \\ \tilde{\tau}_{n+1} = f_{\mathrm{TD}}(|\Re\{R_n(\tau)\}|, \tilde{\tau}_n + (1+\tilde{\alpha}_n T_b), \Delta\tau), \end{cases} \tag{23}$$

where $\hat{\tau}_n$ denotes the time index of peak point of correlation output envelop, and is expressed as

$$\hat{\tau}_k = \max_{\tau_k} |b(k) R_c(\tilde{\alpha}_n(\tau - kT_b))|, \quad k = n, n+1, \tag{24}$$

and $|f_c \Delta\tau| < \pi/2$ is satisfied. Note that for the nth symbol, the correlation output waveform is corresponding to the time index of the peak point of the envelop. For the $n+1$th symbol, the delay estimation of the nth symbol is used to refine the delay estimation result. Moreover, the term of $|\Re\{R_n(\tau)\}|$ is used to eliminate the influence of modulated symbols, i.e., $b(k)$ in (20).

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Then the instant Doppler factor is calculated as

$$\hat{\alpha}_{n+1} = \frac{T_b}{\tilde{\tau}_{n+1} - \tilde{\tau}_n} - 1. \tag{25}$$

C. Demodulation and Doppler Compensation

Recalling the passband cross-correlation outputs in (20), the equivalent baseband signal is expressed as

$$b_n(\tau) = \sum_{k=n}^{n+1} b(k) R_c(\tau - kT_b) \mathrm{e}^{\mathrm{j}2\pi f_c(\alpha_k - \tilde{\alpha}_n)(\tau - \tau_k)}. \tag{26}$$

As shown in (22), the timing of the two symbols are realized through the peak detection, then, the differential encoded symbols are calculated by the baseband signal as

$$\left[\hat{b}_n, \hat{b}_{n+1}\right] = \left[b_n(\tau = \hat{\tau}_n), b_n(\tau = \hat{\tau}_{n+1})\right]. \tag{27}$$

Moreover, the estimation of information symbols can be refined based on the instant estimated Doppler factor as

$$\hat{a}_n = \hat{b}_{n+1} \hat{b}_n^* \mathrm{e}^{-\mathrm{j}\phi_n}, \tag{28}$$

where ϕ_n denotes the residual phase rotation and is expressed as

$$\phi_n = 2\pi(1 + \tilde{\alpha}_n) f_c (\hat{\alpha}_{n+1} - \tilde{\alpha}_n) T_b. \tag{29}$$

D. Local Reference Signal Generator

According to Fig. 2, the correlation output magnitude will diminish caused by the velocity variation, if only the local reference signal is fixed after synchronization, which results in severe SNR loss, and even the correlation peak cannot be distinguished from noise. As shown in Fig. 1, the instant Doppler estimation result is used to generate the local reference signal for the next processing period. It is essential to reduce the Doppler induced magnitude loss, especially for the case that the relative speed between the transmitter and receiver has significant change along with time. However, the local reference signal is not expected to change rapidly. Actually, in most cases, the Doppler spread is changing slowly and slight magnitude loss is tolerable for communication. Moreover, to avoid the instant noise of Doppler estimation, a first-order of a low-pass filter is used to smooth the estimation result, and denoted as

$$\bar{\alpha}_n = \beta \hat{\alpha}_n + (1 - \beta) \bar{\alpha}_{n-1}, \tag{30}$$

where β is a constant coefficient, and satisfies $0 < \beta < 1$. $\bar{\alpha}_0 = \tilde{\alpha}_0$ when $n = 0$. Then the local reference signal generator is denoted as

$$\tilde{\alpha}_n = f_g(\bar{\alpha}_n), \tag{31}$$

where $f_g(\cdot)$ denotes the selection function, which is used to select the stored local reference signal with its corresponding Doppler factor $\tilde{\alpha}_n$ nearest to the filtered instant Doppler factor $\bar{\alpha}_n$.

Table 1 The computational complexity comparison between the proposed method and the CAF method in each processing period

	Proposed method	CAF method
Correlation	$o(2 \times m \mathrm{lb} m)$	$o(2K \times m \mathrm{lb} m)$
Demodulation	$o(2 \times m \mathrm{lb} n)$	$o(2 \times m \mathrm{lb} n)$
Delay estimator	$o(2 \times 1.5\, m)$	–
Doppler estimator	$o(1)$	–
Symbol decision	$o(1)$	$o(2 \times 1.5\, m)$
Ref. sig generator	$o(1.5\, K)$	–

E. Complexity Analysis

The complexity of the proposed algorithm is addressed in this part, and compared with the modified CAF method suitable for DSSS-DBPSK signal. As shown in Fig. 1, the proposed algorithm works in a symbol-by-symbol fashion, and a signal with a duration of two symbols' interval is contained in each processing period. The modified CAF method works in a similar way by correlating the received signal with the whole reference signals in each processing period. Let m denote the length of signal with one symbol interval, K denote the number of local reference signal stored in the memory, and n denote the order of low-pass filter that is used to obtain the baseband signal. It is noticed that $n \ll m$ here. Then, the complexity of the two methods in each processing period is summarized in Tab. 1. The results show that the computation load of the proposed method is lower than the modified CAF method, especially when K is large. Moreover, it is obvious that the correlation module occupies the heaviest computational load in both cases.

IV. Simulation Results

In this section, numerical simulation results of the proposed dynamic Doppler tracking and compensation algorithm are presented. To illustrate the advantages of the proposed algorithm over other Doppler tracking algorithms, the scenario with constant acceleration speed is mainly focused on our simulation. Unless otherwise stated, the symbol interval of the DSSS signal is larger than the maximum multipath spread delay. Therefore, multipath components can be distinguished from each other due to the nearly ideal auto-correlation of the spread sequence. In our simulation, the simplest model of only single path with constant acceleration speed is considered to illustrate the newly dynamic Doppler tracking algorithm.

In our case, for simplifying the computation load and without losing of generality, the spread sequence with a length of 31 and modulation scheme of DBPSK are employed to generate a DSSS signal with a chip rate of 2 500 cps and a center frequency of 12.5 kHz. The signal is sampled with a sampling rate of 100 kHz, and 300 differential encoded symbols are

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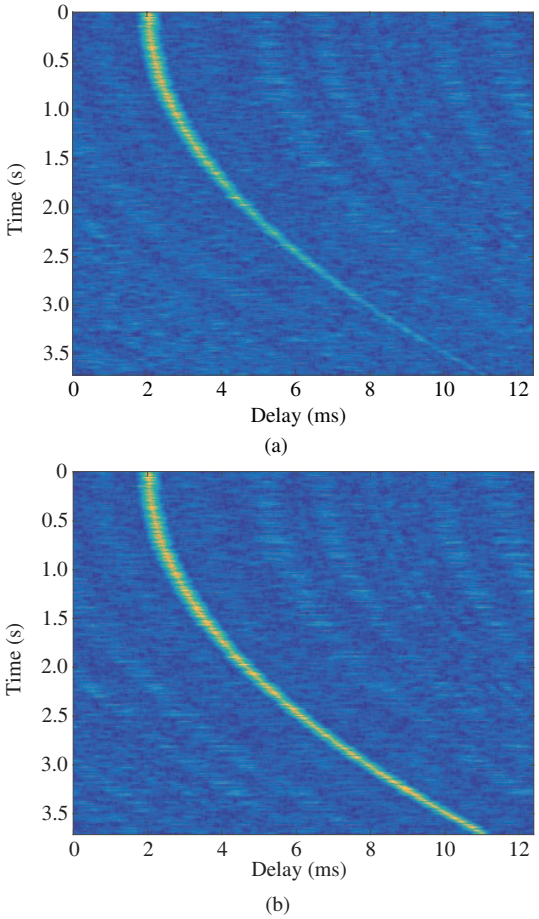

Figure 3 The power-delay-profile of DSSS signals with different methods: (a) the power-delay-profile evolution when the local reference signal is fixed after synchronization; (b) the power-delay-profile evolution when the local reference signal is selected adaptively

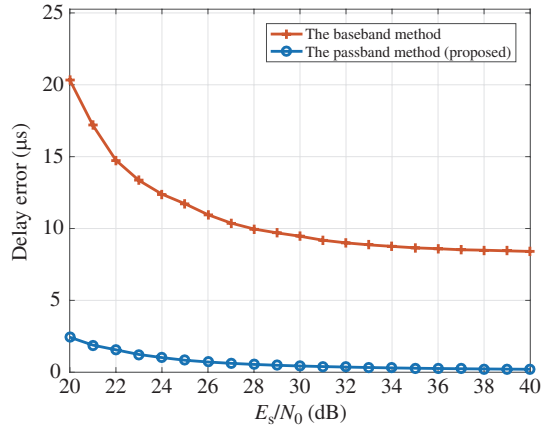

Figure 4 The result of delay estimating resolution versus E_s/N_0

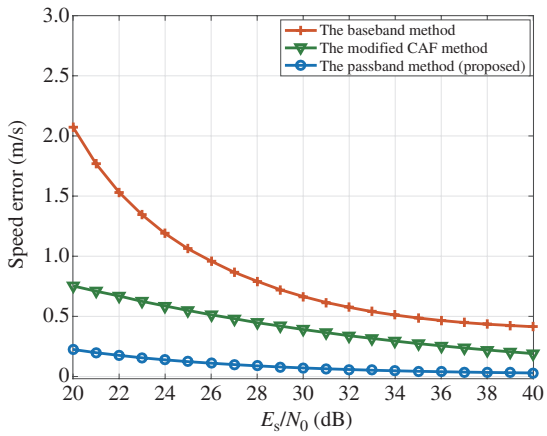

Figure 5 The result of Doppler estimation resolution versus E_s/N_0

transmitted. The duration of transmitted signal is around 3.7 s and the underwater sound speed is assumed to be 1 500 m/s. Assume the transmitter and the receiver are in the same axis, the receiver is fixed and the transmitter is moving away from the receiver with a constant acceleration speed of 1.95 m/s^2, and the initial distance and velocity are 100 m and 0 m/s, respectively.

In the receiver side, the coarse Doppler estimation and compensation is firstly performed, which is usually achieved by calculating the CAF. To simplify our discussion, we assume that the synchronization is ideal and thus the estimated speed is 0 m/s. Here, a total of 11 local reference signals are stored in the memory. The Doppler ranges from -6 m/s to 6 m/s with the step size around 1.2 m/s. Actually, a wider range should be considered in the practical system design. As shown in Fig. 2, the maximum of magnitude loss is less than 0.5 dB when the Doppler step size is set to 1.2 m/s and a length of 31 m-sequence is used. Therefore, the magnitude loss of a DSSS signal is nearly negligible. A denser Doppler step size should be set when the length of spread sequence is longer. The coefficient of first-order low-pass filter for Doppler filtering is set to $\beta = 0.6$.

In Fig. 3, the power-delay-profile evolution as time varies is presented when adjusting the local reference signal adaptively or not. In this case, SNR for each symbol after correlation, i.e., E_s/N_0, is set to 20 dB. The results show that the magnitude of the output envelop decreases along the time axis without dynamic Doppler tracking and local reference signal adjustment, which can be achieved by adopting the dynamic Doppler tracking. The results indicate that the dynamic Doppler tracking is essential for the receiver, especially in the case that the Doppler spread changes significantly during the data packet.

As mentioned above, the kernel behind the proposed dynamic Doppler tracking algorithm is a delay estimator operating on the passband correlation output. The time difference of adjacent symbols is estimated, then the instant Doppler factor is calculated accordingly. In our simulation, two kinds of delay estimator are considered. One is the proposed passband delay estimation method, and the other is baseband delay estimation based on the interpolation for the envelop of the cross-

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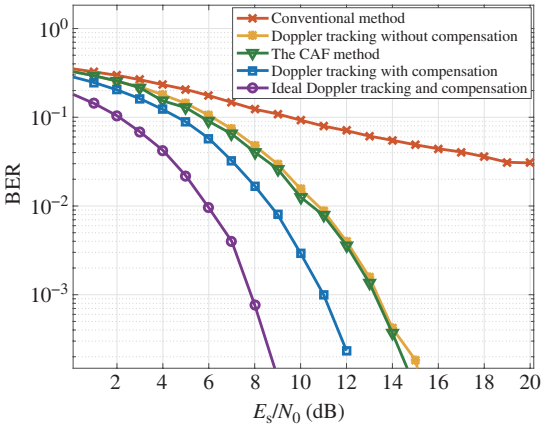

Figure 6 The BER performance of different methods versus E_s/N_0

correlation output. Fig. 4 shows the result of delay estimation resolution versus symbol-level SNR, i.e., E_s/N_0. Compared with the baseband method, the passband method has a definitely accurate delay estimation result. When the E_s/N_0 is equal to 20 dB, the delay estimation resolution of the proposed method is lower than 2.5 μs, i.e., 1/4 sample in this case, while the resolution of the baseband method is higher than 20 μs, i.e., 2 samples.

Fig. 5 shows the result of Doppler estimation error. In addition to the baseband method and the proposed passband method, the modified CAF method suitable for DBPSK is considered here. The basic idea of the modified CAF method is to estimate Doppler spread with different Doppler factors in each processing period. It shows that the Doppler estimation error becomes smaller as E_s/N_0 increases for both methods. The proposed passband method has the highest Doppler estimation resolution compared with the other two methods. When E_s/N_0 equals 20 dB, the velocity estimation error of the proposed passband method is around 0.25 m/s, while the error is more than 2 m/s for the baseband method. When E_s/N_0 equals 40 dB, the velocity estimation error is much lower, around 0.5 m/s for the baseband method, while it is around 0.02 m/s for the proposed method. The simulation result shows that the proposed method is superior than the baseband method. Compared with the delay estimation error results shown in Fig. 4, it is concluded that the Doppler estimation result highly relies on the delay estimation results.

To evaluate the BER performance of the proposed method, 4 different methods are compared in Fig. 6. The first method is a direct decision without both Doppler tracking and compensation, i.e., the conventional method. The second method is the local reference signal which is changing dynamically with Doppler tracking, and does not perform Doppler compensation in the symbol decision part. The third method is the modified CAF method. And the fourth method is the proposed dynamic Doppler tracking and compensation. The 4 methods are all compared with the BER performance of DBPSK when the Doppler effect on the phase and magnitude are ideally compensated.

As shown in Fig. 6, the BER is not equal to zero even when the SNR is high without dynamic Doppler tracking and compensation. It means that the conventional method does not work in the scenario of constant acceleration speed, and the Doppler tracking is required for the receiver. The modified CAF method is to compensate the magnitude distortion induced by Doppler spread, however the phase rotation of residual Doppler spread is not considered. Therefore, it is reasonable that the performance of the modified CAF method is almost the same as the second method that changes local reference signal dynamically without phase compensation as shown in Fig. 6. When the dynamic Doppler tracking is performed, the performance loss is around 5 dB compared with the ideal performance of DBPSK without phase and magnitude loss. When the dynamic Doppler tracking and compensation are performed, the performance loss is around 3 dB compared with the case in AWGN. Another 2 dB performance gain is achieved when the Doppler compensation is integrated with symbol decision. The residual performance loss of the proposed method is mainly due to the mismatch of the received signal and the local reference signal. However, compared with the performance of conventional method, the performance loss of the proposed method is tolerable and it can be further reduced if smaller step size of Doppler is adopted.

V. Conclusion

A symbol-based passband Doppler tracking and compensation algorithm suitable for UWA DSSS signal is proposed in this paper. A newly reception consists of dynamic Doppler estimation and compensation is presented. The kernel of the proposed algorithm is a high resolution of time delay method operating on the passband correlation output. Simulation results show that the proposed method can achieve a Doppler estimation resolution of under 0.3 m/s when the E_s/N_0 is equal to 20 dB. Compared with the ideal DBPSK system, the proposed receiver suffers a performance of around 3 dB, however, the performance is superior to the receiver without dynamic Doppler estimation. Moreover, simulation results also show that dynamic Doppler tracking and compensation is essential to the DSSS signal in the dynamic scenario, especially when the acceleration speed is significantly high.

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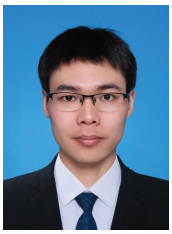

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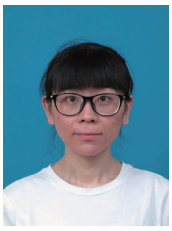

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